CROP AND FERTILIZER ANALYSIS

1. Project Overview and Objective

- ★ This project focuses on cleaning, transforming, and analyzing agricultural crop data using Microsoft Excel.

- ★ The objective is to understand crop cultivation patterns and fertilizer usage across different Crops and Years

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2. Data Sources

Source Description and Timeline: Data government / Agriculture Dataset Search **and** 2020-2025.

- ★ **Domain:** Agriculture Analytics
- ★ **Tool used:** Microsoft Excel , Power Bi

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3. Problem Statement

Analyze crop cultivation patterns and area distribution.

- ★ Evaluate fertilizer usage and quantity applied across crops.
- ★ Compare irrigated and unirrigated crop areas.
- ★ Identify major crops cultivated in different farm sizes.
- ★ Study the relationship between crop area and fertilizer application.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
State	Text	Name of the State
District	Text	District Name
Crop Name	Text	Name of the Crop
Year	Number	Cultivation Year
Nitrogen	Number (Percentage)	Nitrogen Fertilizer
Phosphorus	Number (percentage)	Phosphorous Used
Potassium	Number (Percentage)	Potassium Used

Total Fertilizer	Number (Percentage)	Nitrogen + Phosphorous + Potassium
Total Area	Number	Total Cultivated Area

5. Tools & Technologies

- ★ **Excel:** Data cleaning , Removing blank rows , Creating calculated columns , Creating Pivot Tables for analysis
- ★ **Power BI:** Importing cleaned Excel dataset , Creating charts and visual reports , Adding slicers and filters , building interactive dashboards

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

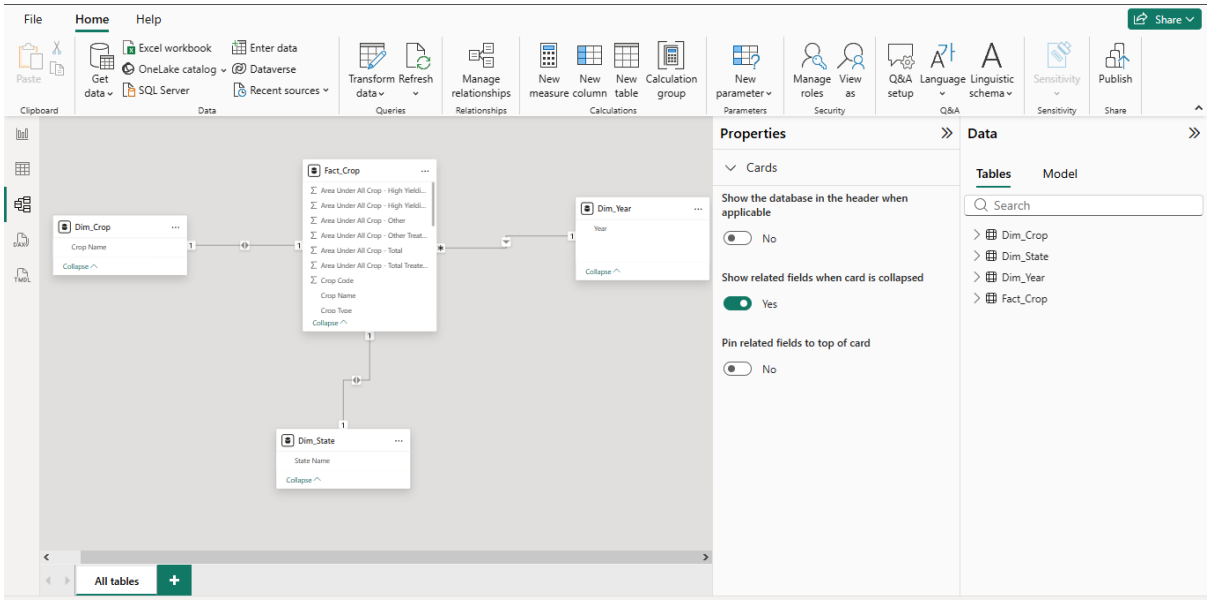
- ★ Checked and confirmed First Row as Header
- ★ Removed blank rows
- ★ Ensured proper number formatting
- ★ Created Calculated Columns:
- ★ Total Fertilizer = Nitrogen + Phosphorus + Potassium
- ★ Total Area = Sum of cultivated area columns
- ★ Converted dataset into structured format

Total Fertilizer%	Total Area
46%	0
46%	817
45%	131
64%	11685
46%	0
25%	30124
64%	0
0%	201
16%	0
64%	1734
57%	0
46%	0
60%	0
46%	7143
60%	274
46%	0
46%	566
20%	361
46%	0
64%	0
55%	0
60%	569
21%	0
62%	0
40%	0
16%	70857
64%	0

7. Data Modelling (Power Bi)

- ★ The dataset was organized into Fact Table and Dimension Tables to structure the data properly for analysis.
- ★ Relationships were created between the Fact table and Dimension tables using common fields such as Crop Name, State, and Year.
- ★ This data model helped in building accurate visualizations and interactive dashboards in Power BI

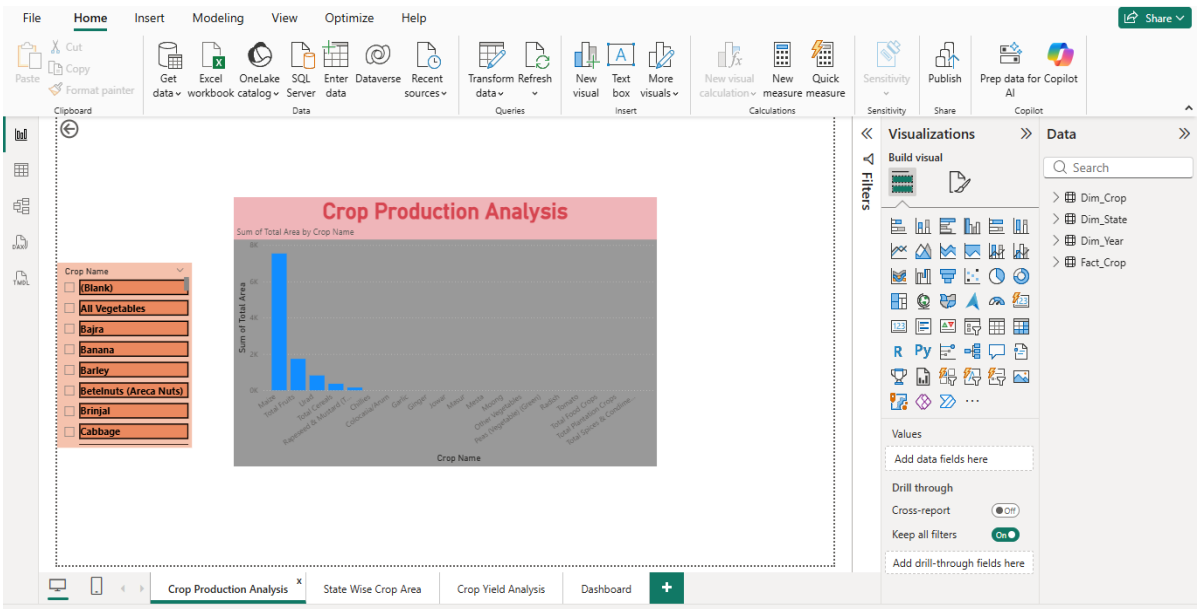
Screenshot of modelling:



8. Analysis and Visualizations (Power BI)

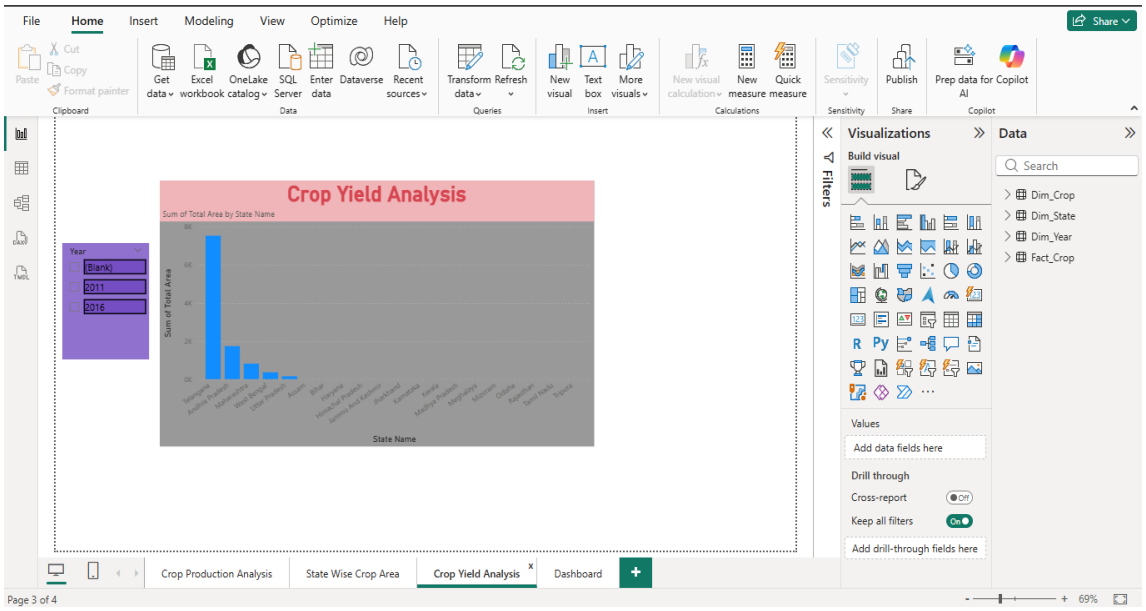
Dashboard Feature:

1.Bar Chart-Crop Production Analysis



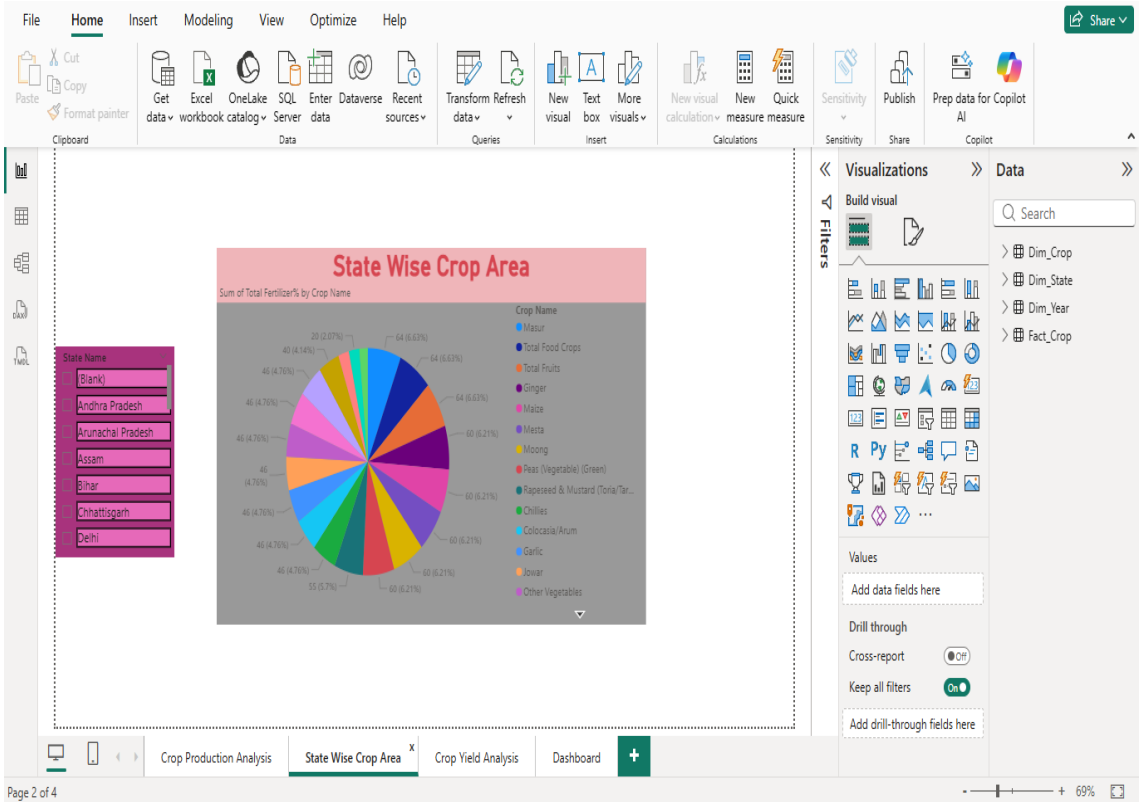
- ★ A bar chart was created to display the total cultivation area of different crops.
- ★ This visualization helps identify which crops occupy larger cultivation areas.

2.Bar chart-Crop Yield Analysis



- ★ A bar chart was created to compare the yield of different crops.
- ★ analysis helps understand the productivity levels of each crop.

3.Pie chart-State Wise Crop Area



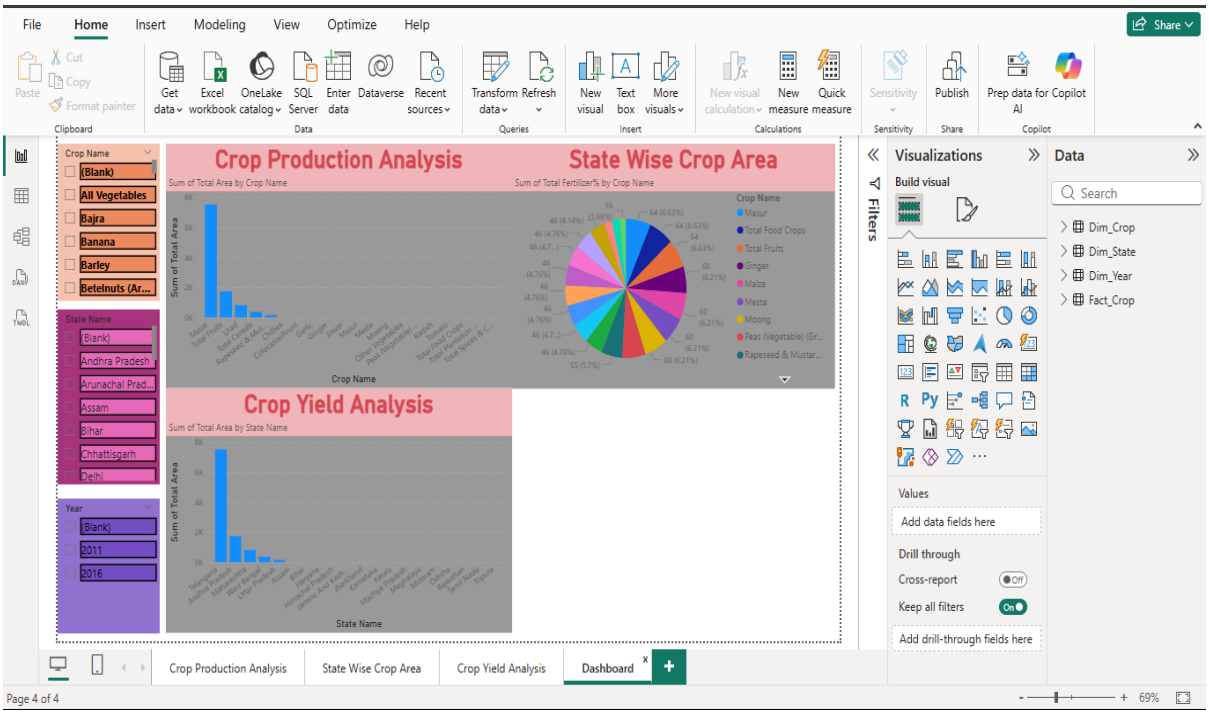
- ★ A pie chart was used to represent the distribution of crop cultivation across different states.
- ★ This helps identify which states contribute more to agricultural production.

4. Interactive Dashboard Filters

Slicers were added to the dashboard to make the report interactive.

The following slicers were used:

- ★ Crop Name
- ★ State
- ★ Year



- ★ These filters allow users to dynamically explore crop production and cultivation pattertns.

9. Insights & Conclusions

- **Key Findings**

- ★ Some crops have larger cultivation areas than others.
- ★ Fertilizer usage is different for each crop.
- ★ Some states produce more crops compared to other states.
- ★ Crop yield varies between different crops.
- ★ Power BI charts help understand crop production data easily

- **Provide the analysis insights:**
  - **Descriptive:** Describes crop production, crop yield, and cultivation patterns using charts and visualizations.
  - **Diagnosis:** Helps identify the differences in crop cultivation and production across crops and states.
  - **Predictive:** Historical agricultural data can help estimate future crop production trends.
  - **Prescriptive:** The insights obtained from the analysis can help improve agricultural planning and resource management.

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10. Conclusions

This project analyzed agricultural crop data using Microsoft Excel and Power BI. The dataset was cleaned and transformed in Excel, and interactive dashboards were created in Power BI to visualize crop production, crop yield, and state-wise cultivation patterns. The analysis helped understand agricultural trends and provided useful insights into crop cultivation and productivity. Data visualization made the information easier to interpret and analyze.